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Quality Analysis and Classification of Rice Grains using Image Processing Techniques

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Abstract - Rice stands as a favored and extensively consumed cereal grain in Asian countries, while also enjoying global accessibility. Within the rice market, the overarching determinant of milled rice lies in its quality, an attribute that assumes heightened significance in the context of import and export trade. Rice samples often harbor assorted extraneous elements such as paddy, chaff, damaged grains, weed seeds, and stones. The principal objective of the proposed approach is to introduce an alternative avenue for quality control and analysis, characterized by reduced expenditure in terms of effort, cost, and time. Image processing emerges as a pivotal and technologically advanced sphere marked by significant advancements. Image processing maneuvers images to execute targeted operations, thereby refining and enhancing the desired outcome. Moreover, this technique enables the extraction of valuable insights from input images. This study strives to develop image processing algorithms with a specific focus on segmenting and identifying rice grains. By harnessing image processing algorithms, it becomes possible to efficiently analyze the quality of grains based on their size. This paper furnishes a solution for the classification and assessment of rice grains, predicated on their dimensions and morphology, through the application of image processing techniques. While prior research has focused on the morphological attributes of grains, encompassing parameters such as area and shape, these endeavors often struggle to yield a generalized formula capable of classifying diverse rice varieties due to the considerable variance in shapes and sizes. In a distinctive departure, this paper augments the analysis by incorporating Fourier features extracted from grain images, thus augmenting the accuracy of classification outcomes.

Key Words: —agriculture, image processing, morphological operations, edge detection, quality analysis, object classification, deep learning, food quality detection

1.INTRODUCTION

The agricultural industry, spanning across centuries, remains expansive and steeped in tradition. The challenge of assessing grain quality has persisted throughout history.

This project introduces a pioneering solution for the evaluation and grading of rice grains by harnessing image processing techniques. Traditionally, the commercial grading of rice hinges on grain size classification, categorizing grains as full, half, or broken. The assessment of food grain quality has conventionally relied on human inspectors employing visual scrutiny. However, the decision-making abilities of human inspectors are susceptible to external influences such as fatigue, subjectivity, and personal biases.

The integration of image processing techniques offers a transformative approach, eliminating the aforementioned challenges while remaining non-destructive and cost-effective. This methodology transcends human limitations, enhancing objectivity and accuracy. The subsequent discussion outlines the procedure deployed to ascertain the percentage quality of rice grains. Rice quality, in essence, is a composite of both physical and chemical attributes. Physical characteristics encompass grain size, shape, chalkiness, whiteness, milling degree, bulk density, and moisture content. On the other hand, chemical attributes involve gelatinization temperature and gel consistency, contributing to the comprehensive assessment of rice quality.

This study centers on the development of image processing algorithms aimed at effectively segmenting and identifying rice grains. The utilization of image processing algorithms proves to be a highly efficient approach for gauging grain quality based on its size. The paper introduces a comprehensive solution for grading and assessing rice grains, focusing on grain size and shape through the application of image processing techniques. Particularly, an edge detection algorithm is employed to discern the boundaries of each grain, employing a technique that identifies the endpoints of individual grains. Subsequently, a caliper is utilized to ascertain the length and breadth of rice grains. This methodology stands out for its minimal time requirement and cost-effectiveness.

In contrast, conventional methods employed for measuring grain shape and size, such as the grain shape tester, dial micrometer, and graphical method, tend to be protracted

and cumbersome. These methods typically allow for the measurement of the dimensions of one grain at a time, yielding results that are not only time-consuming but also susceptible to human errors. Consequently, there is a pressing need for greater precision to fulfill customer expectations and overcome the limitations posed by manual procedures.

Numerous studies have previously delved into the analysis of morphological characteristics of grains, encompassing factors like area and shape. However, the vast diversity in shapes and sizes across different rice varieties precludes the generalization of a uniform formula for classifying all rice types. Addressing this challenge, this paper introduces an additional dimension by extracting Fourier features from grain images, complementing the spatial features and culminating in an elevated level of accuracy for classification purposes.

This paper aims to employ image processing algorithms to analyze grain quality based on size has become a prevalent and effective methodology. This approach facilitates the assessment and classification of rice grain quality by leveraging advanced image processing techniques. By focusing on the dimensions of rice grains, these algorithms contribute to a comprehensive understanding of their quality attributes. This technique holds the potential to revolutionize the conventional methods of evaluating grain quality, providing a more accurate and efficient means of classification.

The remaining part of the paper is organized as follows. Section 2 contains the Literature Survey. The Proposed Model is discussed in Section 3. Section 4 contains the Experiments and Results. Lastly, the Conclusion and Future Directions is presented in Section 5.

2. LITERATURE REVIEW

Food quality detection is a crucial aspect of the food industry, ensuring consumer safety and satisfaction. Recent advancements in machine learning and image processing techniques have revolutionized the accuracy and efficiency of food quality assessment. This literature review aims to provide an in-depth analysis of 15 research papers that explore the integration of machine learning and image processing in food quality detection.

The authors in [1] propose a model to y showcases the application of deep learning techniques, particularly convolutional neural networks (CNNs), for food quality assessment. The authors use image analysis to detect defects, such as mold and discoloration, in food products. The paper[2] focuses on fruit ripeness detection using machine learning algorithms. The authors employ support vector machines (SVM) and random forests to classify fruits into different ripeness categories based on color and texture features. The study[3] presents an automated system for

inspecting bakery products' quality. Image processing techniques are combined with support vector machines for real-time detection of defects and anomalies in baked goods.

The research in[4] focuses on fish quality assessment using image analysis and machine learning. The authors in [5] use features like color, texture, and shape to classify fish into different quality categories, ensuring freshness and safety. This paper introduces texture analysis and neural networks for meat quality detection. Texture features extracted from meat images are fed into neural networks to classify meat products based on tenderness and freshness.

This study [6] explores the use of CNNs for detecting diseases and assessing quality in vegetables. The authors develop a model that can identify diseases and quantify the extent of damage using leaf images. The paper[7] discusses the application of transfer learning and CNNs for food quality inspection. The authors pre-train a CNN on a large dataset and fine-tune it for specific food quality assessment tasks. The authors in [8] focus on contaminant detection in food products using deep learning techniques. The authors train a CNN to identify foreign objects and contaminants, ensuring food safety. This study [9] presents a non-invasive approach to inspect egg quality using machine learning. The authors use image analysis and machine learning algorithms to assess egg freshness and defects.

The paper[10] introduces an automated system for detecting milk spoilage using image processing and neural networks. The authors[11] employ texture and color features to classify spoiled and fresh milk samples. This research focuses on classifying food items based on image features using decision trees. The authors extract color, texture, and shape features to develop a decision tree-based classifier. The study[12] presents a multi-class food quality assessment using deep learning and ensemble methods. The authors in [13] combine the predictions of multiple models to enhance the accuracy of quality assessment. This paper introduces a hybrid CNN-SVM model for quality inspection of fruits. The authors utilize CNN for feature extraction and SVM for classification, achieving improved accuracy in fruit quality assessment. This research[14] employs image processing and random forests for dairy product quality detection. The authors use image features to train a random forest model that identifies defects and anomalies in dairy products. This study focuses on automated detection of freshness in seafood using deep learning techniques. The authors use a deep neural network to assess seafood quality based on color, texture, and shape attributes [15].

In conclusion, the reviewed papers collectively highlight the significant advancements achieved in food quality detection through the integration of machine learning and image processing techniques. From deep learning-based approaches to hybrid models, these studies showcase the potential of technology to enhance food safety, quality, and consumer satisfaction in the food industry.

3. PROPOSED MODEL

Utilizing an image processing technique, the assessment of rice seed quantities is undertaken, followed by their classification based on parameters like length, breadth, and the length-breadth ratio. Specifically, the length represents the average longitudinal dimension of rice grains, while breadth pertains to the average width. The length-breadth ratio is computed as $L/B = [(Average\ length\ of\ rice\ grain)/(average\ breadth\ of\ rice)] * 10$.

The process is delineated through a series of methodological steps:

a. Image Pre-Processing:

The initial phase involves image pre-processing, during which a filter is applied to eliminate noise generated during image acquisition. This filter simultaneously enhances image sharpness. The application of a threshold algorithm aids in segmenting the rice grains from a black background.

b. Shrinkage Morphological Operation:

Subsequently, a shrinkage morphological operation is employed to address the challenge of segmenting touching rice kernels. The process commences with erosion, which effectively separates interconnected features of rice grains without compromising the integrity of individual ones. Dilation follows erosion, with the primary objective being to restore eroded features to their original shape without rejoining previously separated elements.

c. Edge Detection:

Edge detection, the third step, plays a pivotal role in identifying the boundaries of rice grains. The canny algorithm is adopted for its efficiency in detecting edges.

d. Object Measurement:

The fourth stage encompasses object measurement, ascertaining the count of rice grains. Following grain quantification, edge detection algorithms are applied, subsequently yielding endpoint values for each grain. The utilization of a caliper facilitates the connection of endpoints, enabling the measurement of both length and breadth. With these dimensions determined, the length-breadth ratio is calculated.

e. Object Classification:

In the final step of the algorithm, object classification is executed. This necessitates a compilation of standard, measured, and calculated outcomes. Reference data for rice size and shape measurement is sourced from the laboratory manual on rice grain quality, specifically the Directorate of Rice Research located in Rajendra Nagar, Hyderabad.

In conclusion, the systematic application of image processing techniques, encompassing pre-processing, morphological operations, edge detection, measurement, and classification, forms a comprehensive methodology for accurately quantifying and categorizing rice seeds based on their size and shape attributes.

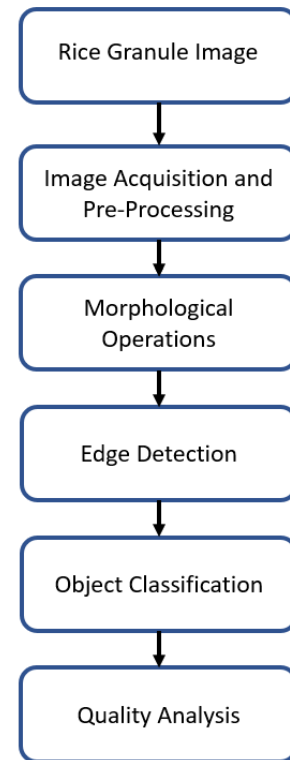


Fig -1: Architecture Diagram of the proposed model

TYPE OF GRANULE	ASPECT RATIO (x)
SLENDER	$2.1 \leq x < 3$
MEDIUM	$1.1 \leq x < 2.1$
BOLD	$0.9 \leq x < 1$
ROUND	

Fig -2: Classification Criteria based on Aspect Ratio

4. EXPERIMENTS AND RESULTS

The primary goal of this project is to minimize the manual labor involved in the classification of rice grains and the assessment of their quality. To accomplish this, the project leverages the power of Machine Learning and utilizes the Python Flask framework. The culmination of the project results in the creation of a functional website. This web application is capable of receiving input in the form of images depicting rice grains. It then employs Machine Learning techniques to effectively categorize the rice grains and simultaneously evaluate their overall quality.

This approach showcases its efficiency not only by reducing the reliance on physical labor but also by providing a cost-effective solution. By amalgamating Machine Learning algorithms with the Python Flask framework, the project successfully streamlines the process of rice grain classification and quality analysis. The end result is a user-friendly website that significantly improves the efficiency and affordability of these critical tasks.

A grouped bar chart is employed in this context to facilitate the classification process. The chart effectively presents information related to the classification of rice grains.

Notably, the chart employs two distinct bars:

1. The blue bar is indicative of the total count of rice grains within the dataset.
2. The red bar, on the other hand, conveys the average aspect ratio of the rice grains.

Furthermore, a pie chart is harnessed for the purpose of quality analysis. This chart serves to visually represent key quality attributes within the sample.

The chart is characterized by the following sections:

1. The blue section of the pie chart denotes the proportion of rice grains present within the given sample.
2. Contrarily, the red section of the pie chart conveys the percentage of dust detected within the analyzed sample.

Both the grouped bar chart and the pie chart play integral roles in conveying vital information regarding the classification and quality assessment of rice grains, respectively. Through visual representation, these graphical elements enhance the comprehensibility and insightfulness of the data analysis process.



Fig -3: Rice Granules Input Image

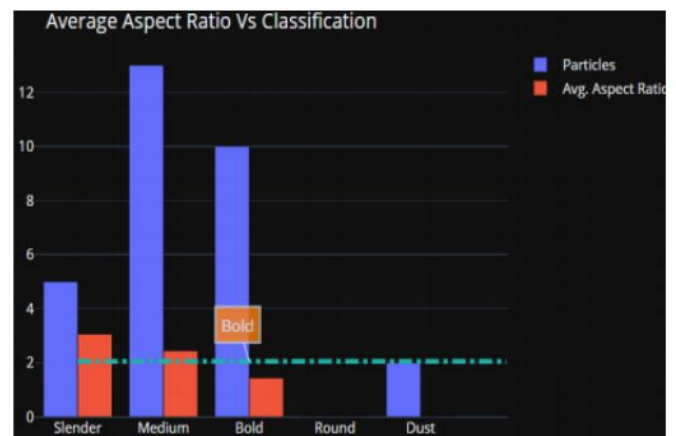


Fig -3: Aspect Ratio vs. Rice Granules Type Classification

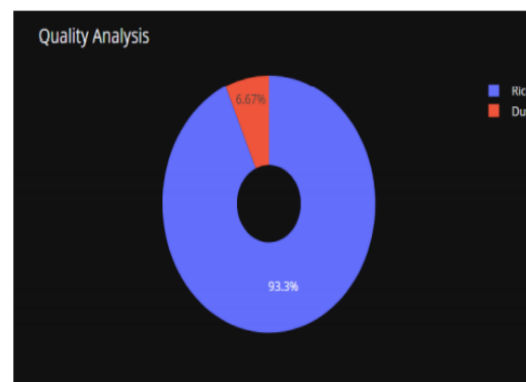


Fig -4: Percentage of Rice and Dust in the rice grains image

5. CONCLUSIONS

In this project, our focus lies in the comprehensive classification of rice grain samples, coupled with a meticulous analysis of their quality based on the aspect ratio. Our approach distinctly differentiates itself from existing works, as it not only identifies rice grains and quantifies their numbers but also delves deeper to evaluate their quality and allocate them to specific categories.

What sets our work apart is its unique capability to achieve near-perfect accuracy in assessing the quality of grains within a sample. This is of paramount importance, particularly for scenarios involving the efficient grading of a large volume of grains. Our methodology significantly expedites this process, alleviating the substantial time and human effort typically associated with manual analysis.

Our image analysis algorithms are applied to images featuring rice grains arranged randomly in a single layer. Addressing potential errors such as touching kernels, our approach utilizes a shrinkage operation to effectively separate interconnected portions. Edge detection is subsequently employed to pinpoint boundary regions and determine the endpoints of each individual grain. Subsequently, using a caliper, we measure the length and

breadth of each grain. These measurements further allow for the calculation of the length-breadth ratio.

Our study is underpinned by the development of image processing algorithms tailored to segment and identify rice grains. The deployment of these algorithms proves highly efficient in evaluating grain quality based on their size. The paramount advantage of our proposed method is its expedited process, minimal time requirement, cost-effectiveness, and superior performance compared to traditional manual methods. All proposed steps have been meticulously executed, culminating in the successful classification and sizing of grains, which are then appropriately categorized according to a predefined table.

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